

The image features the text "DL for ENG" in a bold, 3D, gold-colored font. The letters are floating on a vast, blue ocean with visible ripples and small waves. In the background, there are rolling blue hills or mountains under a sky filled with soft, grey clouds. The overall scene has a serene and expansive feel.

DL for ENG

LECTURE 8.1

Stationary Heat

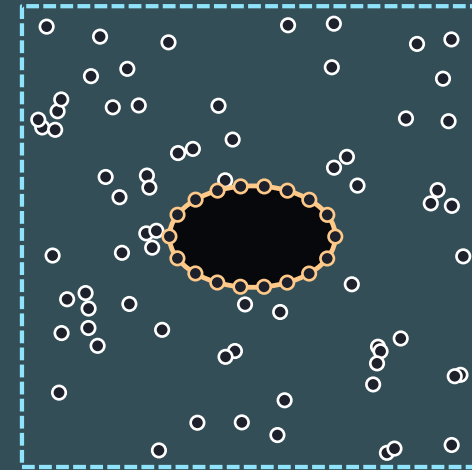
Remus Teodorescu

*With support from Research Assistant Noman Khan -
nomank@energy.aau.dk*

Aalborg University · ret@et.aau.dk

Deep Learning for Engineering · 2026

CONDUCTION · GEOMETRY · THE PLATE



Steady conduction is Poisson's equation with units: Fourier's law, a conductivity, and walls held at a temperature, a flux or a convection. On a plate with a cooling hole the hole is built into the network by its level set, and the answer is checked by an energy balance.

Learning Objectives

SECTIONS, IN ORDER

- 1 **Heat transfer** — conduction, convection, radiation, and which L8 solves
- 2 **Conduction** — the steady equation, variable k , the thermal walls, scaling
- 3 **Geometry** — sampling a domain with a hole, its multiplier, its normal
- 4 **Plate with a hole** — verify, solve, weight, and balance the heat

READ, AND WHAT EACH ANSWERS

Incropera ch. 1	how does heat move?
Liu ch. 2.4	what does a thermal problem need?
Liu ch. 3	how is a domain with a hole handled?
Liu ch. 4.6	how is the answer checked?

Recap

1 – 2 · STEADY

1 What decides the type of a second-order equation, and which conditions does each type need?

How it treats time. An elliptic equation has none: a condition on every wall. A parabolic one has one time derivative: the walls and one initial field. A hyperbolic one has two: the walls and two initial conditions. With fewer, the problem has many answers. *Liu ch. 2; Ex_07.2 notebooks 1 – 3*

2 What do Poisson's and Helmholtz's equations describe, and what must a Helmholtz network be protected from?

Poisson is a steady balance between a source and diffusion. Helmholtz adds a term in the field itself, so it has modes, and $u \equiv 0$ always satisfies it: the network needs a unit-norm term and a floor on λ . In the slot, λ reaching the first eigenvalue is thermal runaway. *Liu ch. 2.4, 6.2; Ex_07.2 notebook 2*

3 – 4 · IN TIME

3 What does time add: how do the heat and the wave equation treat a disturbance, and an error made early?

Heat spreads and forgets: every feature decays, the sharpest fastest, and an early error is damped. A wave travels at a finite speed and keeps its shape, so an early error is carried along. Heat needs one initial condition, the wave two. *Liu ch. 4.1, 5.1; Ex_07.2 notebooks 1, 3*

4 How does a PINN meet the conditions of each type, and which type is hardest for it?

A condition with a formula is built into the trial function: the walls, the initial field. One without, like the panel's initial velocity, is a term in the loss. The wave is hardest: its error is not damped, and its residual must be scaled before the network moves. *Liu ch. 4.2, 5.2; Ex_07.2 notebook 3*

Conduction

Conduction — heat carried through a material by its atoms and electrons, with no motion of the material itself. It flows from hot to cold, at a rate proportional to the temperature gradient: Fourier's law, with the conductivity k as the property of the material.

FOURIER'S LAW

$$\mathbf{q} = -k \nabla T$$

$\mathbf{q}$ the heat flux in W/m^2 , k the conductivity in $\text{W}/(\text{m K})$, ∇T the gradient in K/m

- **driven by** — a temperature gradient, inside a solid or a still fluid
- **direction** — from hot to cold, hence the minus sign
- **examples** — a chip into its heat sink, a winding into its slot

CONDUCTION IN PYTORCH

```
q = -k * grad(T, xy)          # Fourier's law: flux from the gradient
```

CONDUCTIVITY OF COMMON MATERIALS

material	k , $\text{W}/(\text{m K})$
copper	400
aluminium	237
silicon	150
stainless steel	15
glass	1
water	0.61
air	0.03

Typical values at room temperature. Four orders of magnitude lie between copper and air: a heat sink is metal, and insulation is trapped air.

Convection

Convection — heat carried away from a surface by a moving fluid. The motion of the fluid is not solved for: its effect is put into one coefficient h , and the flux is proportional to the difference between the surface and the fluid. That is Newton's law of cooling.

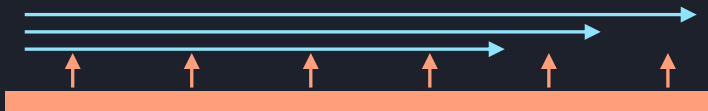
NEWTON'S LAW OF COOLING

$$q = h (T_s - T_\infty)$$

h the heat transfer coefficient in $\text{W}/(\text{m}^2 \text{ K})$, T_s the surface, T_∞ the fluid far from it

- **natural** — the fluid moves by itself: warm fluid rises
- **forced** — a fan or a pump moves it, and h is larger
- **examples** — a heat sink in air, a cold plate with water

CONVECTION, DRAWN



the fluid, at T_∞ , moves past

the surface, at T_s , gives up q

HEAT TRANSFER COEFFICIENT

flow	h , $\text{W}/(\text{m}^2 \text{ K})$
natural, air	2 – 25
forced, air	25 – 250
natural, water	50 – 1 000
forced, water	100 – 20 000
boiling water	2 500 – 100 000

Typical ranges. h is not a property of the fluid alone: it depends on the speed and on the shape, and comes from a correlation or a measurement.

Radiation

Radiation — heat exchanged as electromagnetic waves between surfaces that see each other; it needs no material in between. The power a surface radiates grows with the fourth power of its absolute temperature: the Stefan–Boltzmann law, with the emissivity ε of the surface.

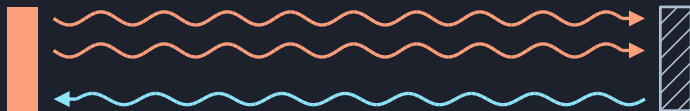
STEFAN–BOLTZMANN LAW

$$q = \varepsilon \sigma (T_s^4 - T_{\text{sur}}^4)$$

ε the emissivity, 0 to 1, $\sigma = 5.67 \times 10^{-8} \text{ W}/(\text{m}^2 \text{ K}^4)$, T_{sur} the surroundings

- **needs** — no medium: it crosses a vacuum
- **nonlinear** — the fourth power, with T in kelvin
- **examples** — the sun, a furnace wall, a radiator in space

RADIATION, DRAWN



the hot surface radiates

the surroundings radiate back

EMISSIVITY OF COMMON SURFACES

surface	ε
polished aluminium	0.05
anodised aluminium	0.80
black paint	0.95
a black body	1

Typical values. A surface at 80 °C with $\varepsilon = 0.90$, in surroundings at 25 °C, radiates 390.50 W/m². Natural convection at $h = 10 \text{ W}/(\text{m}^2 \text{ K})$ carries 550 W/m² from the same surface: the two are of one size.

Heat Transfer in L8

Heat transfer in L8 — of the three mechanisms only conduction is an equation for a field inside the solid, and that field is what a PINN solves for. Convection and radiation act at the surface: they enter as boundary conditions, and their own physics is left out.

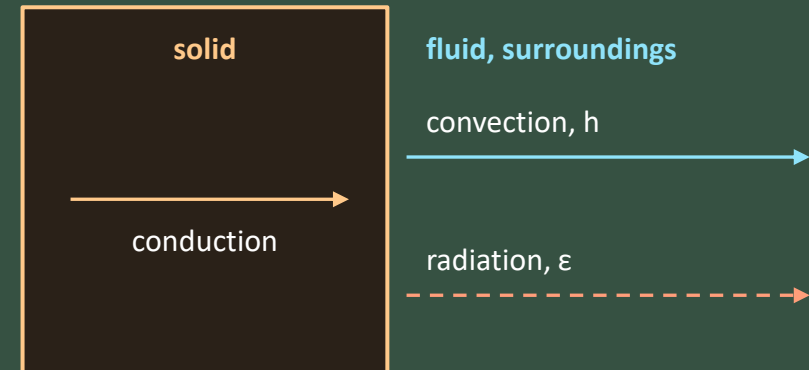
WHERE EACH MECHANISM ENTERS

	acts	in L8
conduction	inside the solid	the equation the PINN solves
convection	at a surface, into a fluid	a Robin wall, with h given
radiation	between surfaces	left out, or counted in h

WHY THE OTHER TWO ARE NOT SOLVED

- **convection** — its field is the flow of the fluid: that is L9
- **radiation** — T^4 and the view between surfaces: nonlinear
- **both** — act at the wall, so a boundary condition carries them

ONE BODY, THREE MECHANISMS



Radiation is not small: at 80 °C against 25 °C, with $\varepsilon = 0.90$, it acts like a coefficient of 7.10 W/(m² K), the size of h in still air. Where it matters, L8 counts it in h ; it does not solve it.

Steady Conduction

Steady conduction — Fourier's law in an energy balance: in steady state what leaves a small volume equals what is generated in it, $\nabla \cdot \mathbf{q} = Q$. With $\mathbf{q} = -k \nabla T$ that is Poisson's equation for the temperature, with Q/k as its source.

FOURIER'S LAW IN AN ENERGY BALANCE

$$\mathbf{q} = -k \nabla T, \quad \nabla \cdot \mathbf{q} = Q$$

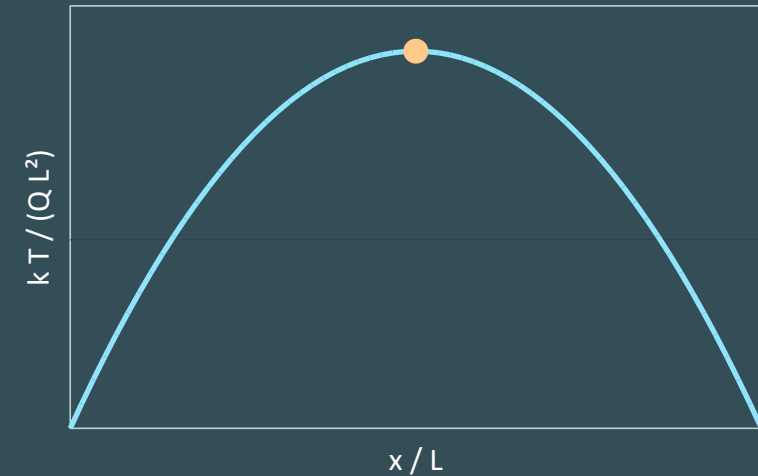
$$T_{xx} + T_{yy} + \frac{Q}{k} = 0$$

$\mathbf{q}$ the heat flux, k the conductivity in W/(m K)
 Q the heat generated in W/m³

CONDUCTION IN PYTORCH

```
T = net(xy)                # the temperature, in kelvin
q = -k * grad(T, xy)        # Fourier's law; grad = autograd
F = div(q, xy) - Q          # the energy balance, zero inside
```

A WALL THAT GENERATES HEAT



Both faces held at T_0 . The profile is a parabola, and at the amber peak the gradient, and so the flux, is zero: heat leaves through both faces.

Variable Conductivity

Variable conductivity — when k changes with position it stays inside the divergence: $\nabla \cdot (k \nabla T) + Q = 0$. Across an interface between two materials the gradient jumps and the normal flux does not, so the divergence must not be expanded; autograd differentiates $k \nabla T$ as written.

K INSIDE THE DIVERGENCE

$$\nabla \cdot (k(\mathbf{x}) \nabla T) + Q(\mathbf{x}) = 0$$

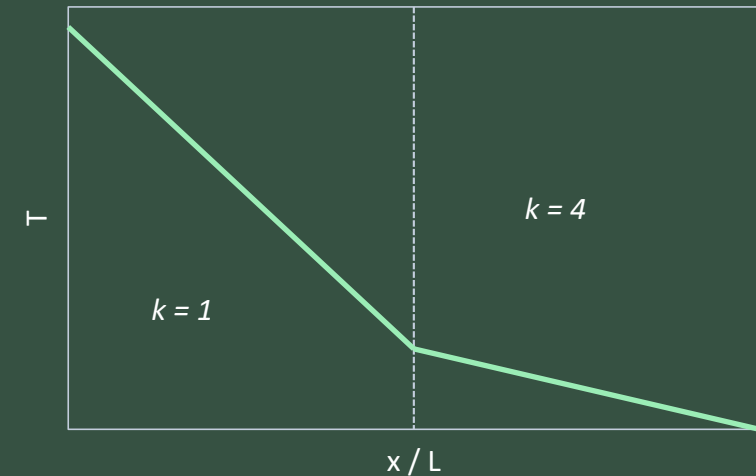
$k(\mathbf{x})$ the conductivity at each point, $k \nabla T$ differentiated as one quantity

- **at an interface** — the slope jumps, the flux does not
- **in code** — differentiate the product, not its parts

VARIABLE k IN PYTORCH

```
k = k_field(xy)           # the conductivity at each point
q = -k * grad(T, xy)       # Fourier's flux, one quantity
F = div(q, xy) - Q         # differentiated as written
```

TWO MATERIALS, ONE INTERFACE



Left face at 1, right face at 0, no generation. The slope falls fourfold at the interface; the flux $k dT/dx$ is the same on both sides.

Thermal Boundary Conditions

Thermal boundary conditions — three ways to fix a wall: a temperature (Dirichlet), a heat flux (Neumann), or convection to a fluid (Robin), which ties the flux to the wall temperature. Only a temperature can be built into the network; the other two stay loss terms.

A TEMPERATURE, A FLUX, A CONVECTION

Dirichlet

$$T = \bar{T} \text{ on } \partial\Omega_D$$

Neumann

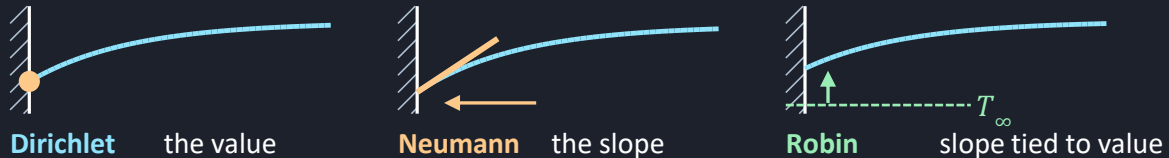
$$-k \frac{\partial T}{\partial n} = \bar{q} \text{ on } \partial\Omega_N$$

Robin

$$-k \frac{\partial T}{\partial n} = h(T - T_\infty)$$

$\bar{T}$ the wall temperature, $\bar{q}$ the wall heat flux, n the outward normal
 h the heat transfer coefficient, T_∞ the fluid temperature

WHAT EACH CONDITION FIXES AT THE WALL



ROBIN · CONVECTION AT THE WALL

- **balance** — heat conducted to the wall equals heat the fluid carries away
- h — heat transfer coefficient, $\text{W}/(\text{m}^2 \text{ K})$, or film coefficient: the fluid layer at the wall in one number
- **radiation** — where it matters, it is counted in h

BIOT NUMBER

$$\text{Bi} = \frac{hL}{k} = \frac{L/k}{1/h}$$

L the size of the body, L/k resists conduction inside
 $1/h$ resists convection at the wall

$\text{Bi} \ll 1$: nearly insulated (Neumann). $\text{Bi} \gg 1$: the wall is at T_∞ (Dirichlet). 1 mm of silicon in still air: 1.67×10^{-4} .

Well-Posedness

Compatibility condition — a problem with a flux condition on every wall has a solution only if the heat out through the walls equals the heat generated, and then only up to a constant: any shift of T fits the same equation and walls. A PINN picks one without warning.

HEAT OUT EQUALS HEAT GENERATED

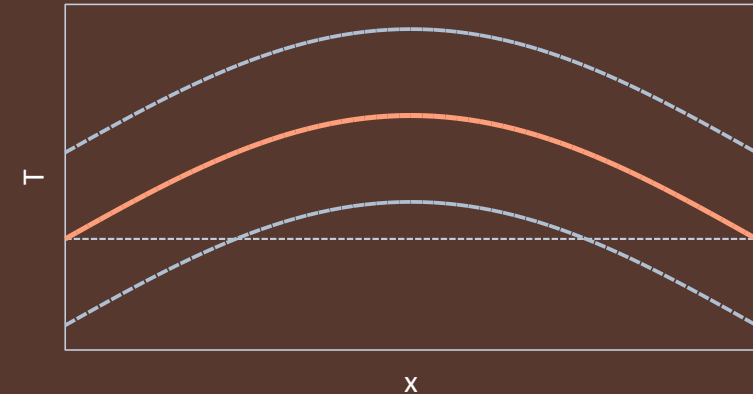
$$\oint_{\partial\Omega} \bar{q} ds = \int_{\Omega} Q d\Omega$$

$\bar{q}$ the flux out through the boundary $\partial\Omega$, Q the heat generated inside

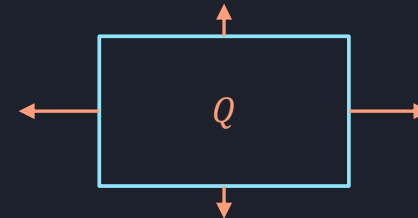
- **one value pinned** — fixes the constant
- **one wall at a temperature** — fixes it too

TALK · 2 MIN How would you explain why a problem with only flux conditions has no single solution, in your own words? Then, in the plot on the right, which of the three curves is right, and what would you add to the loss to choose it?

THE SAME FIELD, SHIFTED



THE BALANCE, DRAWN



heat generated inside
= heat out through the walls
*Only then is there a steady field,
and it can still be shifted up or
down.*

Non-Dimensional Form

Non-dimensional form — the problem rewritten in scaled variables: lengths divided by the device size L , temperature by a rise $\Delta T = QL^2/k$. Every term is then of order one, as the initialisation and the loss weights assume. Train scaled, and convert back at the end.

SCALE, TRAIN, CONVERT BACK

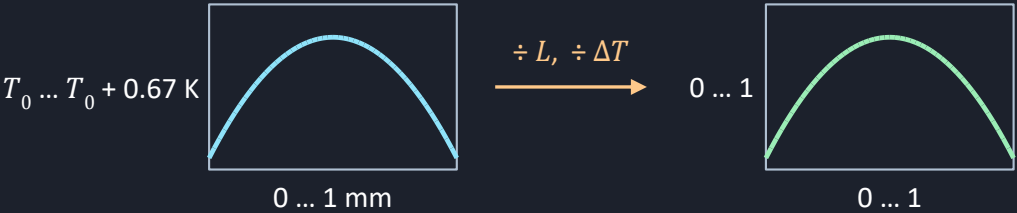
$$\theta = \frac{T - T_0}{\Delta T}, \quad \xi = \frac{x}{L}, \quad \Delta T = \frac{Q L^2}{k}$$

θ the scaled temperature, ξ the scaled coordinate, ΔT the temperature scale

$$\theta_{\xi\xi} + \theta_{\eta\eta} + 1 = 0$$

$\eta = y/L$; the same equation, every coefficient equal to one

THE SAME CURVE, SCALED



A SILICON DEVICE, BEFORE AND AFTER

	physical	scaled
length L	1 mm	1
conductivity k	150 W/(m K)	1
source Q	10^8 W/m ³	1
rise ΔT	$Q L^2/k = 0.67$ K	1
coefficient h	25 W/(m ² K)	$Bi = 1.67 \times 10^{-4}$

In metres and kelvin the source term is ten to the eighth and the length ten to the minus third. Scaled, every term is of order one.

Sampling a Real Domain

Rejection sampling — points drawn uniformly over a box around the domain, keeping those the level set places inside, $\phi > 0$. On a curved wall the points are spaced by arc length rather than by angle, so the ends of the hole, where it bends most, get their share.

ONE FORMULA: INSIDE, OUTSIDE, ON THE WALL

$$\phi(x, y) = \left(\frac{x - x_c}{a}\right)^2 + \left(\frac{y - y_c}{b}\right)^2 - 1$$

(x_c, y_c) the centre, a, b the half axes; $\phi < 0$ in the hole, $\phi = 0$ on its wall

- **inside** — keep a point where $\phi > 0$
- **on the wall** — equal steps of arc length

THE LEVEL SET ALONG A LINE THROUGH THE HOLE

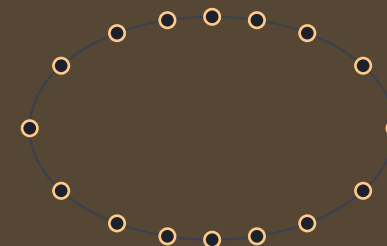
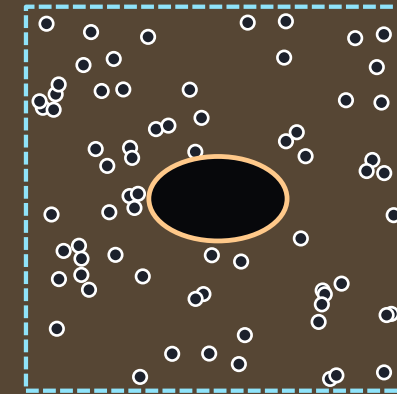


$\phi > 0$: in the plate, kept

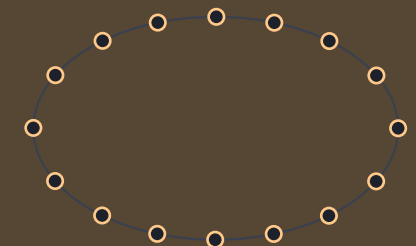
$\phi = 0$: the hole's wall

$\phi < 0$: in the hole, rejected

THE PLATE, AND ITS WALL POINTS



equal angle: ends bare



equal arc length

Boundary Multiplier

Boundary multiplier — a function that is zero on a boundary and positive in the domain. Multiplied into the network's output, it makes a value hold there exactly. For an ellipse the level set ϕ is one; for a shape with no formula, a small second network is trained to be one, then frozen.

THE HOLE WALL, BUILT IN

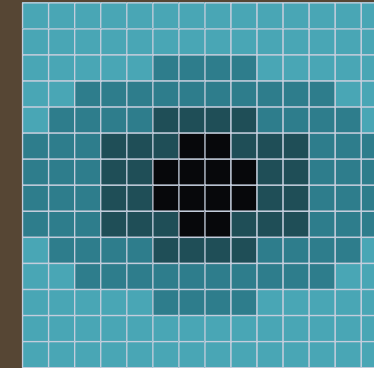
$$T(\mathbf{x}) = \phi(\mathbf{x}) \mathcal{N}_{\hat{w}}(\mathbf{x})$$

ϕ the level set, zero on the hole; $\mathcal{N}_{\hat{w}}$ the network; $T = 0$ there for any weights

- **a value** — held exactly by a factor
- **a derivative** — cannot be, stays a loss term

TALK · 2 MIN How would you explain what a boundary multiplier does, in your own words? Then, in the figure on the right, where is the multiplier small, and what does that do to the network's gradient near the hole?

ϕ OVER THE PLATE



dark: the hole

FROM THE WALL INTO THE PLATE



Flux on a Curved Boundary

Normal derivative — the rate of change of T across a wall, $\partial T / \partial n = \nabla T \cdot \mathbf{n}$, where $\mathbf{n}$ is the unit normal pointing out of the material. On a curved wall $\mathbf{n}$ changes with position: it comes from the gradient of the level set, and its sign decides whether heat leaves or enters.

THE NORMAL FROM THE LEVEL SET

$$\frac{\partial T}{\partial n} = \nabla T \cdot \mathbf{n}, \quad \mathbf{n} = -\frac{\nabla \phi}{|\nabla \phi|}$$

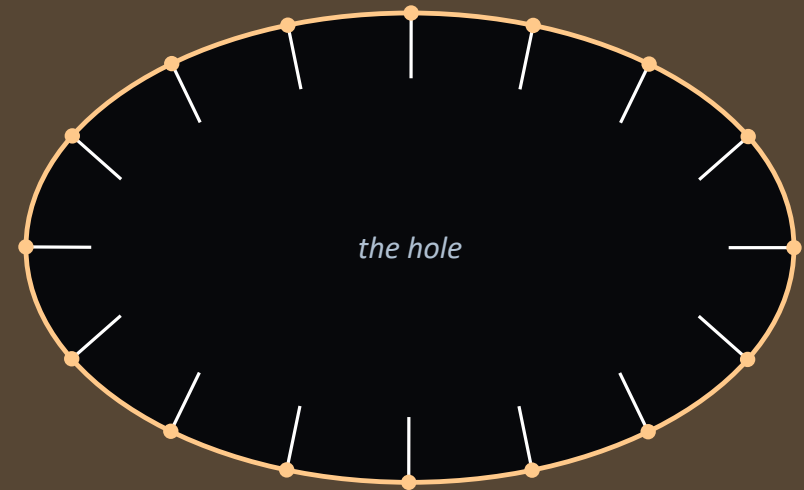
$\mathbf{n}$ the unit normal out of the plate; minus, because out of the plate is into the hole

- **on an edge** — the normal is a constant
- **on the hole** — it turns with the wall

THE NORMAL IN PYTORCH

```
g = grad(phi(xy_h), xy_h)           # the level set's gradient
n = -g / g.norm(dim=1, keepdim=True) # unit normal, out of the plate
q_n = -k * (grad(T, xy_h) * n).sum(1) # Fourier's flux through the wall
```

NORMALS ON THE HOLE WALL



Each white stick is $\mathbf{n}$ at an amber point: it turns with the wall and always points out of the plate. A sign error inverts the flux, not the magnitude.

Manufactured Solution

Manufactured solution — a solution chosen in advance, with the source derived from it by differentiation, so the answer is known exactly. Solving that problem checks the residual, the trial solution and the training loop before they meet a problem with no answer to compare against.

CHOOSE THE ANSWER, DERIVE THE SOURCE

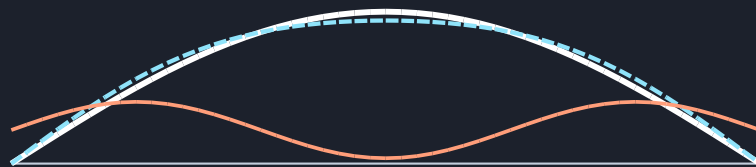
$$T = \sin(\pi x)\sin(\pi y) \Rightarrow \frac{Q}{k} = 2\pi^2 \sin(\pi x)\sin(\pi y)$$

T the chosen answer, zero on the four edges; Q/k the source derived from it

$$\varepsilon = \frac{\|T_{\hat{w}} - T_{\text{exact}}\|_2}{\|T_{\text{exact}}\|_2}$$

$T_{\hat{w}}$ the network's answer, T_{exact} the chosen one

THE ANSWER IS KNOWN, SO THE ERROR IS A CURVE

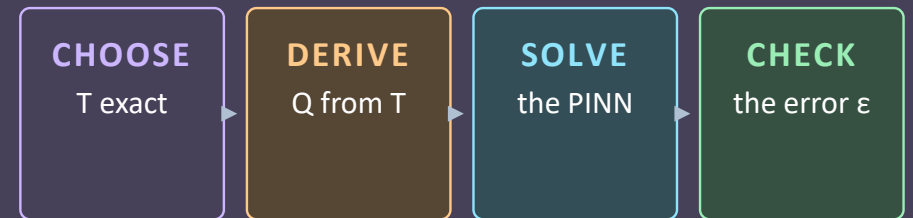


the chosen answer

the network's

their difference, enlarged

FOUR STEPS BEFORE THE PLATE



what it checks

the residual, the trial, the loop

what it cannot

the geometry of the plate

An error found here costs minutes; the same error found on the plate costs an afternoon.

Heated Plate with a Hole

Heated plate with a hole — the case study of this block: a square plate with uniform heat generation, cooled through an elliptical hole held at $T = 0$, with insulated outer edges, $\partial T / \partial n = 0$. The hole wall is built into the network by its level set; the edges stay a loss term.

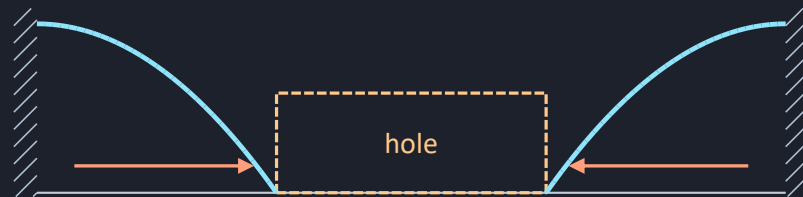
ONE BUILT-IN WALL, ONE LOSS TERM

$$\mathcal{L} = \mathcal{L}_{PDE} + w_N \mathcal{L}_{NBC}$$

$\mathcal{L}_{PDE}$ the residual inside, $\mathcal{L}_{NBC}$ the insulated edges, w_N their weight

- **the hole** — $T = 0$, by the multiplier
- **the edges** — no flux, a weighted term

THE TEMPERATURE ALONG A LINE THROUGH THE HOLE

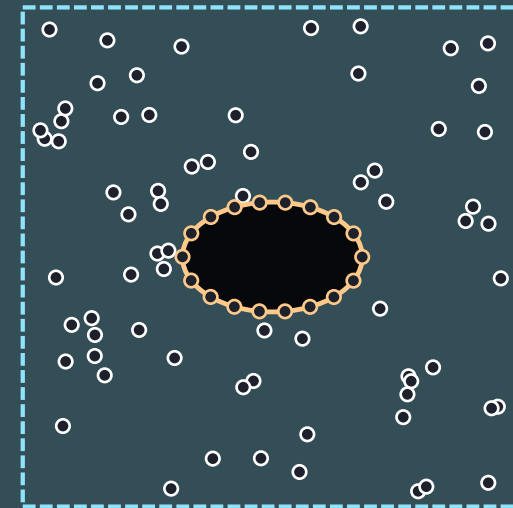


insulated edges: flat

hole wall: $T = 0$, built in

the heat runs to the hole

THE PLATE OF EX_08.1



amber wall

$T = 0$, built in

dashed edges

$\partial T / \partial n = 0$, a loss term

Flux Weight

Flux weight — the weight w_N on the insulated-edge term. Too small, and the network ignores the edges and heat leaks through them; too large, and the residual inside pays for a perfect wall. It is chosen by a sweep that watches both terms, not the total loss.

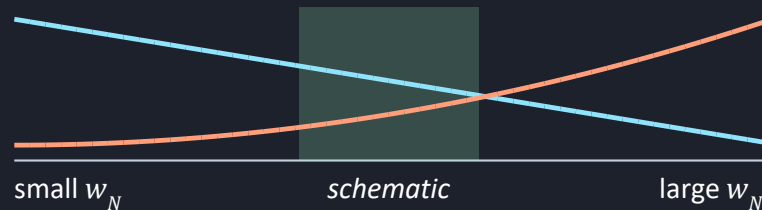
ONE RUN PER WEIGHT, BOTH TERMS WATCHED

$$\mathcal{L} = \mathcal{L}_{PDE} + w_N \mathcal{L}_{NBC}$$

$\mathcal{L}_{PDE}$ the residual inside, $\mathcal{L}_{NBC}$ the insulated edges, w_N their weight

- **the total loss** — hides which term is paying
- **the choice** — a trade, argued from both

ONE TERM FALLS, THE OTHER RISES



the edge term: falls

the residual inside: rises

between: argued from both

WHAT EACH END OF THE SWEEP DOES

w_N small

the edges leak heat

w_N large

the residual inside pays

between

argued from both terms

The weight is a rate of exchange between a flux and a residual, which have different units. Scaling changes the rate before any sweep does.

Energy Balance Check

Energy balance check — a check on the trained network, not a new law. Fourier's law gives the heat flux at each point of the hole's wall; added up around the hole, that is the heat leaving the plate. In steady state it equals the heat generated: the energy balance, for the whole plate.

FOURIER'S FLUX, ADDED UP AROUND THE HOLE

$$P_h \overline{q_n} = Q A_\Omega$$

P_h the hole's perimeter, $\overline{q_n}$ the mean of Fourier's flux into it, A_Ω the plate's area

- **at a point** — Fourier's law gives the flux into the hole
- **around the hole** — its mean times the perimeter is the heat out
- **in the plate** — Q times the area is the heat generated

THE BALANCE IN PYTORCH

```
q_n = -k * (grad(T, xy_h) * n).sum(1)    # 1: Fourier's flux at each wall point
out = q_n.mean() * perimeter              # 2: heat out; points evenly spaced
gen = Q * (1 - pi * a * b)                # 3: heat generated; compare with out
```

THE CHECK IN EX_08.1

	value
heat generated	9378 W/m
heat out	9332 W/m
heat out / generated	99.51 %
out short by	0.49 %

The plate with a hole has no exact solution to compare against, so this balance is a check with a physical reason to come out right: heat generated is known, heat out is the network's. Per metre of depth. Whether 0.49 % is small enough is your decision.

Advanced Topics

RELATED TOPICS LEFT OUT, AND WHERE THEY MATTER

topic	what it adds	read	where it makes a difference
energy method	an energy, first derivatives only	Liu ch. 2.8	elasticity, structures
triangulated domain	points and weights from a mesh	Liu ch. 3.2	any drawn geometry
convergence study	error against points and size	Liu ch. 4.9.3	a design sign-off
<u>diffusion surrogate</u>	trained on solved cases, then steered	UDL ch. 14 – 18	many boundary conditions
PINN or FEM	when each is the better tool	Liu ch. 2.10	one-off steady solves

WHY THEY ARE LEFT OUT

The plate of Ex_08.1 needs none of them: its hole has a formula, its answer can be checked by an energy balance, and it is solved once. A finite-element solver would solve it too; the method earns its place on the inverse problems of L8.2.

Exercise 8.1

ONE NOTEBOOK, SECTION BY SECTION

- **1 – 3** — the plate, its data, its physics, scaled
- **4** — a fitted reference, and finite differences
- **5** — a PINN with the hole built in; the three compared
- **6** — what the comparison says
- **7 – 8** — the report; mini projects MP8.1A and MP8.1B

THE QUESTION EACH PART ANSWERS

- 1 – 3** how is the plate posed?
- 4** what is the answer, to 0.10 K?
- 5** how does a PINN compare?

REPORT QUESTION, TO CONCLUDE

For this one plate, solved once, which of the three methods would you choose, and what would have to change for the network to be worth its training?

Key Takeaways

HEAT TRANSFER

what it is	conduction, convection, radiation
function	T by conduction; the rest are walls
train, tune	scale first; flux walls weighted
vs Poisson	units, k inside, three walls
used for	devices, plates, heat sinks

REAL GEOMETRY

what it is	a domain with no polynomial factor
function	ϕ : inside, multiplier, normal
train, tune	arc length on the wall; w_N
vs rectangle	n and the factor vary with x
used for	plates with holes, device sections

Next — L8.2: the same plate, in time.

Questions

1 – 2 · HEAT TRANSFER

1 What are the three mechanisms of heat transfer, and which of them does a PINN solve in L8?

Conduction through a material (Fourier's law), convection from a surface into a moving fluid (Newton's law of cooling, with h), and radiation between surfaces (Stefan–Boltzmann, T^4). Only conduction is a field equation inside the solid; the other two act at the surface and enter as boundary conditions. *Incropera ch. 1*

2 What does a steady conduction problem need before a PINN solves it, and how is the result checked?

Fourier's law in an energy balance gives Poisson's equation, with k inside the divergence. Each wall carries a temperature, a flux or convection; with a flux on every wall the level is not fixed. Scale it so every term is of order one, then check by a manufactured solution and the energy balance. *Liu ch. 2.2, 2.4, 4.4; Ex_08.1 sections 3 – 5*

3 – 4 · REAL GEOMETRY

3 How is a domain with a hole sampled, and how is a temperature built into its wall?

Points are drawn over a box and kept where the level set ϕ is positive; on the curved wall they are spaced by arc length. A boundary multiplier, zero on the wall and positive inside, multiplies the network's output, so the wall temperature holds exactly. *Liu ch. 3.1 – 3.3; Ex_08.1 section 5*

4 How is a flux condition imposed on a curved wall, and how is its weight chosen?

As a loss term on the normal derivative, the gradient of T along the outward unit normal, which comes from the level set. Its weight is found by a sweep that watches the flux term and the residual separately: too small and heat leaks through the wall, too large and the interior pays. *Liu ch. 2.2.3, 3; Ex_08.1 section 5*

NEXT

L8.2 — Transient Heat

The same plate, in time: an initial field, causality, and a diffusivity recovered from sensors.

BEFORE THE NEXT SESSION

read	Liu ch. 3, 4.6
run	Ex_08.1, sections 1 to 6
bring	the energy balance, and your error
think about	which of your walls can be built in

The steady plate is solved and checked. Next it heats up.